

Week 6: Evaluation Layer — Semantic Proxy + Regression Taxonomy

Auto-generated 2026-05-21 15:49 from outputs/week06_bertscore_proxy/ and outputs/week06_analysis/. Do not edit by hand. Re-run python -m src.reporting.build_report --week week06 to refresh.

1. Objective

W3 (paired BM25-vs-reranked generation) and W5 (full-dev rerank) gave us four surface-form deltas (ROUGE-L, BLEU, EM, Token-F1) with paired-bootstrap CIs that all exclude zero. W6 asks two follow-up questions that the surface-form table cannot answer:

1. Does the rerank Δ also show up in a *semantic* similarity metric, or are we just rewarding lexical overlap?
2. The full-dev split has 6,980 paired queries; 233 are regressions where the reranker brings the relevant passage into top-3 yet the generator scores *worse* than under BM25. What breaks on those queries — retrieval, semantics, or generation?

Both questions are answered cheaply (CPU, ~minutes) and produced under outputs/week06_*/. Neither requires re-running W3/W5.

2. Setup

2.1 BERTScore semantic-proxy

- Inputs: the same two paired prediction files used in the W3 surface-form CI (outputs/week03_generation_bm25_full/predictions.jsonl and outputs/week03_generation_reranked_full/predictions.jsonl).
- Subsample: **3,000 of 6,980 shared qids**, seed 42 (deterministic).
- Scorer: **distilbert-base-uncased** with rescale_with_baseline=True, multi-reference max-F1 per query.
- CI: paired bootstrap, 10,000 resamples, seed 42, 95 % percentile interval.

This is deliberately *not* the canonical roberta-large BERTScore. It is a fast proxy whose only purpose is to answer "is the surface-form Δ also visible in a semantic-similarity scorer?". The full-dev roberta-large pass is listed under §7 *Next*.

2.2 Regression failure taxonomy

- Pool: the **233 regression queries** identified by `scripts/analyze_generation_rerank.py` (relevant passage in top-3 under rerank, but generator F1 dropped vs BM25).
- Sample: **40** seeded queries, seed 42.
- Labels are produced by a deterministic rule cascade (`scripts/regression_failure_taxonomy.py`); first-match wins: `truncation_short`, `truncation_midword`, `topic_drift`, `extractive_passage_bias`, `semantic_mismatch`.

This is a *triage*, not a citation-grade classifier — the value is in the headline split, not the per-row labels.

3. BERTScore proxy results

Metric	BM25 (mean)	Reranked (mean)	Δ (rerank – BM25)	95 % CI	p (two- sided)
BERTScore-F1 (DistilBERT proxy)	0.2192	0.3920	+0.1728	[+0.1608, +0.1850]	< 0.001

Per-query split on the 3,000-pair subsample:

Outcome	Rate
Rerank strictly better	64.8 %
Tie	5.7 %
BM25 strictly better	29.5 %

The semantic Δ (+0.1728) lands within a hair of the surface-form Token-F1 Δ already published in the W3 paired-bootstrap table (+0.1711) and the ROUGE-L Δ (+0.1742). **The rerank gain is not a surface-form artefact** — a semantic-similarity scorer picks it up at the same magnitude.

4. Regression taxonomy

40-query seeded triage of the 233-strong regression bucket.

Failure mode	Count	% of sample
<code>truncation_midword</code>	22	55.0 %
<code>truncation_short</code> (≤ 3 tok)	14	35.0 %
<code>topic_drift</code>	2	5.0 %
<code>extractive_passage_bias</code>	2	5.0 %
<code>semantic_mismatch</code>	0	0.0 %

90.0 % of regressions are generation-side truncation, not retrieval or semantic failures. The cross-encoder gets the relevant passage into the generator's window; the generator then emits a short or mid-sentence answer. This *looks* like the `max_new_tokens=64` cap firing; a follow-up run at `max_new_tokens=128` falsifies that reading — see §6 *Closure of the budget-cap hypothesis*.

`semantic_mismatch` did not pick up any sampled rows under the rule cascade. Per the script's documentation that bucket is the residual catch-all; its emptiness here is consistent with the BERTScore result in §3 (the rerank gain is real on a semantic scorer).

5. Limitations

- **DistilBERT, not roberta-large.** The headline $\Delta + \text{CI}$ in §3 is a *proxy*. For citation in a cross-paper comparison, re-run `scripts/bertscore_paired_eval.py` with `--model-type roberta-large --n-pairs 0` (full dev, ~hours of CPU). The proxy is calibrated only for "is the surface-form Δ also visible in a semantic scorer", not for absolute BERTScore numbers.
- **Sampled, not full-dev BERTScore.** $3,000/6,980 = 43.0\%$ of the shared qid set; full-dev is trivially obtainable but does not change the qualitative answer.
- **Heuristic taxonomy.** The five labels in §4 are deterministic rules, not a learned classifier. Labels are intentionally coarse; the load-bearing claim is the truncation share, not the per-row attribution.
- **Single sampled triage.** 40 of 233 = ~17 % of the regression bucket. Seed-stable, but a different seed would shift the exact percentages by a few points.

6. Closure of the budget-cap hypothesis

The §4 taxonomy at the time predicted that raising `max_new_tokens` from 64 $\rightarrow$ 128 would close most of the regression bucket. A follow-up sweep on full dev/small (same generator, prompts, retrieval inputs, seed; CLI override only, canonical 64-token outputs untouched) **falsifies** that prediction:

signal	mnt=64 (canonical)	mnt=128
Token-F1 Δ (rerank – BM25)	+0.1711	+0.1706
ROUGE-L Δ	+0.1742	+0.1736
BLEU Δ	+0.1330	+0.1325
EM Δ	+0.0471	+0.0471
Regression bucket size	233	231
Truncation share (40-query triage)	90.0 %	87.5 %

All four surface-form deltas move by <0.005 ; the regression bucket shrinks by 2 queries; the truncation share moves 2.5 percentage points — all within noise on a 40-query sample. T5-small is hitting EOS naturally on this prompt format, not running out of decode budget. The mid-word-ending output style is intrinsic to the model.

Outputs under `outputs/week03_generation_{bm25, reranked}_full_mnt128/`, `outputs/week06_bootstrap_mnt128/`, `outputs/week06_taxonomy_mnt128/` (all gitignored). BERTScore proxy on the new predictions was skipped and remains optional; see *Reproduce — closure run* below.

7. Next

- **Generator-side work targets capacity / output style, not decode budget.** The closure result above re-aims any further work on the generator: instruction-tuned or QA-style small models, or moving off T5-small entirely. The decode budget is no longer a candidate bottleneck.
- **Full-dev BERTScore with roberta-large.** Convert the §3 proxy into a citation-grade evaluation. Same script, two CLI flags.
- **Failure-mode classifier upgrade.** The residual regression bucket — once budget is ruled out — is dominated by topic-drift / extractive-bias / semantic-mismatch; a learned classifier (rather than the current rule cascade) is appropriate at that point.

7.1 W6 follow-ups (done; live in the README)

After this snapshot was written, three offline analyses were layered on top of the same per-query metrics. They do **not** require any new generation / reranker run; numbers are not embedded here to keep the W6 snapshot stable, see the README §1 Week 6 for the live writeup.

- *W6-A* — question-form tagging of dev/small queries (`scripts/tag_query_forms.py`, output `outputs/week06_querytype/`).
- *W6-B* — rerank Δ by question-form, with paired-bootstrap CIs per form (`scripts/analyze_rerank_by_query_form.py`, output `outputs/week06_rerank_by_form/`). Headline: which is the only form whose Δ Token-F1 CI includes zero.
- *W6-C* — Mann-Whitney comparison of regression vs non-regression queries on five structural features (`scripts/regression_query_profile.py`; boxplots under `figures/`). Result: regressions look structurally identical to the rest on the query side; only the retrieved top-3 passage length is slightly shorter ($p = 0.0073$, $r = -0.103$).

8. Reproduce

```
# (1) Generation on full dev/small – done in W3, but listed here
# for
# completeness; mutually restricted to the shared qid set:
python experiments/run_generation_baseline.py \
  --input-run outputs/week02_bm25/run.tsv \
  --output-dir outputs/week03_generation_bm25_full \
  --retrieval-source bm25 \
  --restrict-to-run outputs/week05_reranker_full/run.tsv \
  --num-eval-queries 9999

python experiments/run_generation_baseline.py \
  --input-run outputs/week05_reranker_full/run.tsv \
```

```

--output-dir outputs/week03_generation_reranked_full \
--retrieval-source reranked \
--restrict-to-run outputs/week02_bm25/run.tsv \
--num-eval-queries 9999

# (2) Bucket / retrieval-flag analysis – emits regression bucket
and
# summary.json under outputs/week06_analysis/:
python scripts/analyze_generation_rerank.py \
--bm25-dir outputs/week03_generation_bm25_full \
--reranked-dir outputs/week03_generation_reranked_full \
--output-dir outputs/week06_analysis

# (3) BERTScore semantic-proxy CI (~3 min CPU on the 3 000-pair
subsample):
python scripts/bertscore_paired_eval.py \
--bm25-dir outputs/week03_generation_bm25_full \
--reranked-dir outputs/week03_generation_reranked_full \
--output-dir outputs/week06_bertscore_proxy \
--n-pairs 3000

# (4) Regression failure taxonomy (40-query seeded triage):
python scripts/regression_failure_taxonomy.py

```

Reproduce — closure run (max_new_tokens=128)

The sweep and the two cheap analyses; BERTScore on the new predictions is optional and was skipped at write time.

```

# (1) Full BM25-vs-reranked sweep at max_new_tokens=128. Writes to
# fresh _mnt128 directories; the canonical _full outputs are not
# touched.
python scripts/run_full_generation_and_analysis.py \
--max-new-tokens 128 --out-suffix _mnt128

# (2) Paired-bootstrap CI on the new predictions:
python scripts/bootstrap_generation_comparison.py \
--bm25-dir outputs/week03_generation_bm25_full_mnt128 \
--reranked-dir outputs/week03_generation_reranked_full_mnt128
\
--output-dir outputs/week06_bootstrap_mnt128

# (3) Regression failure taxonomy on the new predictions:
python scripts/regression_failure_taxonomy.py \
--bm25-dir outputs/week03_generation_bm25_full_mnt128 \
--reranked-dir outputs/week03_generation_reranked_full_mnt128
\
--per-query outputs/week06_analysis_mnt128/
per_query_metrics.jsonl \
--output-dir outputs/week06_taxonomy_mnt128

# (4) Optional: BERTScore proxy on the new predictions (~15–30
min):

```

```
python scripts/bertscore_paired_eval.py \  
  --bm25-dir outputs/week03_generation_bm25_full_mnt128 \  
  --reranked-dir outputs/week03_generation_reranked_full_mnt128 \  
  \  
  --output-dir outputs/week06_bertscore_proxy_mnt128 \  
  --n-pairs 3000
```